



Exchange rate elasticities of international tourism and the role of dominant currency pricing

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ABSTRACT

In this paper, we estimate exchange rate elasticities of international tourism. Both the bilateral exchange rate and the U.S. dollar exchange rate relative to tourism origin countries are important drivers of international tourism flows. The U.S. dollar exchange rate is more important for tourism destination countries with higher U.S. dollar borrowing, pointing to a complementarity between U.S. dollar pricing and financing. Country-specific dominant currencies (CSDCs) play only a minor role on average but are important for tourism-dependent countries and those with a high concentration of foreign tourists across origin countries. Consistent with the literature of dominant currency pricing (DCP), we also find that local hotel prices do increase strongly when the domestic currency depreciates against the U.S. dollar. The importance of the U.S. dollar exchange rate represents a strong piece of evidence of DCP in the international trade of services. The results suggest that the benefits of exchange rate flexibility for tourism-dependent countries may be weaker than previously thought and that a broad appreciation of the U.S. dollar can be associated with a significant decline in tourism flows globally.

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1. Introduction

The extent to which a flexible exchange rate can act as a shock absorber is one of the most debated topics in the international trade literature. Under the traditional Mundell-Fleming framework, an exchange rate depreciation should have a positive effect on export volumes. This traditional Mundell-Fleming framework, however, has been called into question by the recent development of the dominant currency pricing (DCP) literature. The extensive use of the U.S. dollar in setting prices for international trade, regardless of the origin or the destination of trade flows, can mute the reaction of export volumes to exchange rate movements (Gopinath, 2015; Goldberg and Tille, 2008; Gopinath et al., 2020a). While the DCP literature has presented strong evidence of U.S. dollar pricing in the international trade of goods, empirical evidence on the sensitivity of services flows to exchange rate movements is scarce.

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In this paper, we zero in on one important service sector—international tourism—to shed light on the exchange rate elasticities of services trade and prices. International tourism is an important growth engine for many countries in the world—the United Nations World Tourism Organization (UNWTO) data suggests that spending by international tourists is around 5% on average globally, but contributes up to 25% for some tourism-dependent countries. Furthermore, by our estimation, spending on hotels on average accounts for 60% of the total expenses for international tourists, pointing to the importance of the hotel industry especially in tourism-dependent economies and hence the need to better understand the exchange rate implications on hotel pricing and arrivals. By studying a large sample of international tourism flow data, we provide strong evidence that both bilateral exchange rate movements and orthogonal U.S. dollar movements are important drivers of international tourism flows. Moreover, U.S. dollar movements significantly affect hotel prices in local currency.

Quantitatively, we show that when the origin-country currency depreciates relative to the destination-country currency by 10 percent, bilateral tourism flows decline by 1.1 percent without controlling for how the origin-country currency fluctuates against the U.S. dollar. Once we control for the U.S. dollar exchange rate, the elasticity of the bilateral tourism flows is reduced to 0.7 percent in response to a 10 percent depreciation. At the same time, a 10 percent depreciation of the origin-country currency against the U.S. dollar reduces bilateral travel flows by 1.1 percent (controlling for the bilateral exchange rate movements). As an example, not only an appreciation of the British pound relative to the Mexican peso, but also an appreciation of the British pound relative to the U.S. dollar, would increase the number of tourists traveling from the UK to Mexico. These results are robust across a wide range of countries regardless of exchange rate regimes.

The strong effect of the U.S. dollar in driving tourism demand is at odds with implications of the conventional literature in which exporters are assumed to set prices of exports purely in producer currencies (PCP), and export volumes respond positively to domestic currency depreciation. Rather, our results are consistent with findings of the DCP literature, suggesting that cross-border tourism products may be partly priced in the U.S. dollar regardless of the origin or the destination of tourists. On average, both the bilateral exchange rate and the U.S. dollar exchange rate are important drivers of tourism flows. However, for some small tourism-dependent economies, if a large share of its hotels set their prices in the U.S. dollar and these prices are sticky, foreign demand for hotels in this country (and exports of tourism in general) may decline when a strengthening of the U.S. dollar makes the local hotels more expensive for potential foreign tourists, regardless of the movements of the bilateral exchange rates. In that sense, an appreciation of the broad U.S. dollar index could lead to a decline in tourism inflows.

To further test this hypothesis, we complement our tourism flow (quantity) analysis with an analysis of the pricing of hotels. If hotel prices were purely set in domestic currencies, we would not expect hotel prices (in the local currency) to systematically change in the short-term relative to the fluctuation of the exchange rate against the U.S. dollar. Using a large sample of hotel price data, we find that local hotel prices do increase strongly when the domestic currency depreciates against the U.S. dollar. We estimate an average pass-through from the U.S. dollar movements to local hotel prices of 0.4, indicating rather strong U.S. dollar pricing for hotels across the world.

The average price and quantity elasticities mask significant heterogeneity across countries. One potential explanation for the cross-country variation in DCP is the complementarities between U.S. dollar pricing and financing (Gopinath and Stein, 2020). When companies borrow in the U.S. dollar, pricing their exports in the dollar provides a natural hedge against movements in the dollar exchange rate (Casas et al., 2020). We test for those complementarities by exploiting variation in the degree of U.S. dollar indebtedness across countries. Our results show that the role of the U.S. dollar is significantly stronger for countries where a relatively large share of corporate borrowing is in the U.S. dollar, suggesting that depending on the liability dollarization of a country, either the bilateral exchange rate or the U.S. dollar exchange rate can be the main driver of tourism flows.

Another potential explanation for the cross-country variation may stem from the country- or region-specific characteristics of the composition of foreign tourists. While countries in the Caribbean may choose to invoice their hotels in the U.S. dollar given the proximity and the large share of U.S. tourists for the local markets, small tourism-dependent countries close to Europe may choose to use the euro. To capture this effect, for each tourism destination country, we define a “country-specific dominant currency,” which we dub CSDC, as the currency of the country where the largest share of tourists resides. We then test for its importance in driving tourism flows. On average, we do not find strong evidence that the CSDC plays a large role in driving tourism flows, controlling for other exchange rate movements. However, the role of the CSDC gains weight for destination countries where tourist arrivals are highly concentrated and for those that have relatively high tourism reliance. These results suggest that the composition and intensity of tourist arrivals may affect the invoicing decisions by local tourism sectors.

The methodology in estimating exchange rate elasticities at the country-pair level follows closely the DCP literature (i.e., Gopinath et al., 2020a). Instead of focusing solely on the movements of bilateral exchange rates, we estimate the role of the U.S. dollar (and other potential dominant currencies) regardless of the tourist source or destination countries. By using bilateral tourist arrival data, we can control for time-varying destination-specific shocks in the regressions and exploit the heterogeneity in exchange rate movements among different currency pairs that may affect the volume of tourist arrivals.

A comparison with the existing DCP literature can help put our analysis into perspective. For international trade of goods, Gopinath et al., 2020a estimated the bilateral exchange rate elasticity to be 0.03 and the U.S. dollar elasticity to be 0.19. Our estimated elasticity for international tourism is 0.07 for the bilateral exchange rate and 0.11 for the U.S. dollar exchange rate. In other words, the U.S. dollar exchange rate plays a quantitatively more important role than the bilateral exchange rate for

both international trade of goods and tourism, while DCP, on average, is stronger for international trade of goods than for tourism.

By using high-quality data on both prices and quantities, our work also provides a unique example to disentangle the various currency pairs in analyzing exchange rate elasticities. In comparison, comprehensive granular cross-country data on both prices and quantities are often unavailable in the international trade literature, which is why many papers have to focus on individual countries (e.g., [Gopinath and Rigobon, 2008](#); [Fitzgerald and Haller, 2012](#)). Even when cross-country quantity data are available, the heterogeneity in international trade of goods makes cross-country or sector comparisons difficult, forcing researchers to focus on individual goods instead (e.g., [Chen and Juvenal, 2016](#)). On the other hand, the hotel price and tourist arrival (by country of origin) data that we use in this paper allow us to focus on international tourism, which arguably represents a homogenous product in international trade.

Our findings also fill a gap in the economics literature of international tourism, which has focused mostly on the bilateral exchange rate movements between two countries and their effect on tourist flows. For example, [Gray \(1966\)](#) and [Vilasuso and Menz \(1998\)](#) study the income and exchange rate elasticities of the demand for travel between the United States and Canada. [Chandra, Head, and Tappata \(2014\)](#) analyze the decision to travel across international borders for Canada–U.S. travel using micro-level data and showed that an appreciation of the home currency increases outbound travel. Similarly, [Neiman and Swagel \(2009\)](#) find that a stronger dollar (a real depreciation of the currency of the origin country) leads to less travel to the United States. [De Vita \(2014\)](#) investigates the impact of exchange rate regimes on international tourism using a sample of 27 high-income countries and showed that maintaining a relatively stable exchange rate is important to attract international tourist arrivals. [Gopinath et al., \(2020b\)](#) also study service sector trade and analyze tourist arrivals in the European Union. In contrast, in this paper we study a variety of exchange rate elasticities for tourism quantities and shed light on the pricing of hotels in response to exchange rates for a large share of countries. To our best knowledge, our paper is the first in the literature to quantify exchange rate elasticities for prices and quantities in a setting with multiple exchange rate pairs.

Our work also contributes to the growing literature on dominant currencies. The empirical work by [Goldberg and Tille \(2008\)](#) and [Gopinath \(2015\)](#) first found that international trade tends to be invoiced in a small number of “dominant currencies,” especially the U.S. dollar. [Gopinath and Rigobon \(2008\)](#) and [Fitzgerald and Haller \(2012\)](#) further demonstrated that international trade prices tend to be rigid in such currencies. The DCP framework proposed by [Gopinath et al., 2020a](#) demonstrates that the extensive use of a third country’s currency (such as the U.S. dollar) in setting prices for international trade, regardless of the origin or destination of trade flows, can dampen the short-term reaction of export volumes to exchange rate movements. We contribute to this literature by providing evidence that DCP is prevalent not only in goods trade, but also in international tourism. And our results indicate that the benefits of domestic currency depreciation in order to boost tourism exports may be weaker than previously thought under the traditional Mundell–Fleming framework.

Finally, our findings have important policy implications for small tourism-dependent economies. Many of these countries are among the hardest hit by the COVID-19 pandemic because of their heavy reliance on exports of tourism ([Ferretti and Maria, 2021](#); [Gorreti et al., 2021](#)). Understanding how exchange rate movements are driving tourism flows is particularly important, as these economies have limited policy options to regain competitiveness in the post-pandemic market. In the Caribbean, for example, tourism contributes a large share to the overall economy, both directly and indirectly. While many of these countries adopt a fixed exchange rate with currencies pegged to the U.S. dollar, the movements between the dollar vis-à-vis the currencies of other major tourist source countries (e.g., Canada, the European Union, and the United Kingdom) could have a direct impact on competitiveness, both within the Caribbean and versus destinations in other regions that provide similar tourism products. A strengthening of the U.S. dollar could render the currency peggers in the Caribbean less attractive for non-U.S. tourists, other things being equal. Even for countries with a flexible exchange rate (e.g., the Dominican Republic and Jamaica), a depreciation of the domestic currency vis-à-vis the U.S. dollar does not necessarily improve competitiveness if their accommodation services, which typically account for the lion’s share of the domestic value added of the tourism industry, are invoiced in the dollar. In other words, the role of exchange rate as a shock absorber may be limited because of DCP.

The rest of the paper is organized as follows. Section II introduces the data used in the paper and provides summary statistics. In Section III, we present evidence on elasticities of hotel prices with respect to exchange rate movements. Section IV quantifies the elasticity of tourist arrivals with respect to movements of exchange rates between various currency pairs. In Section V, we draw the connection between DCP and dominant currency financing (DCF). In Section VI, we present evidence on the importance of CSDCs for certain countries. Section VII provides evidence on how the share of domestic tourism shapes exchange rate elasticities. Section VIII concludes.

2. Data and summary statistics

For the quantity of international tourism, we use the bilateral data from the United Nations World Tourism Organization (UNWTO). The data are based on the annual outbound tourism data (trips abroad by residents to destination countries) for the period of 1995 to 2019. UNWTO compiles this information using data provided by each destination country. The data set includes various types of outbound tourism, such as arrivals of nonresident tourists at national borders by nationality and by country of residence, arrivals of nonresident visitors at national borders by nationality and by country of residence, arrivals of nonresident tourists in hotels and similar establishments by nationality and by country of residence, and arrivals of non-

resident tourists in all types of accommodation establishments by nationality and by country of residence. Due to data completeness and appropriateness for our analysis, we use the arrivals of nonresident tourists at national borders by country of residence for our main analysis. Hence, we only consider tourists and not business travelers, as the latter are likely to be less responsive to exchange rate movements. We also use country of residence instead of nationality as the relevant metric, as the former is more representative of the demand side of international tourism. Our final data set covers 181 destination and 200 origin countries from 1995 until 2019 at an annual frequency.²

For the price of international tourism, we use data provided by Tripadvisor (<https://www.tripadvisor.com>). The data set consists of the annual average daily rates (in U.S. dollar) of 6,500 hotels in 61 countries for 2014 to 2019 based on users' search results on the Tripadvisor website.³ In our main analysis, we aggregate the data to the country-year level. The Tripadvisor "bubble" ratings of the hotels based on users' ratings and reviews (from one bubble to five bubbles, with one meaning "terrible" and five meaning "excellent") also allow us to split the sample based on the ratings to have a more granular analysis on the exchange rate elasticities, as the higher-rated hotels usually correspond to the more expensive ones. In what follows we label 1–3 bubbles as "Low Quality", 2–3 as "Medium Quality", and 4–5 as "High Quality".

The U.S. dollar debt data are taken from Adler et al. (2020). The overall measure is available for 36 major advanced economies and emerging market economies for 2001 to 2019. The corporate foreign currency exposure is constructed by adding foreign currency corporate debt securities (from the Bank for International Settlements (BIS) International Debt Statistics), cross-border foreign currency loans to nonfinancial firms (from BIS Locational Banking Statistics), and local foreign currency loans to nonfinancial firms (from the International Monetary Fund (IMF) Monetary and Financial Statistics). Finally, we use the exchange rate data from the IMF International Financial Statistics. We trim all data on the 1st and 99th percentile to avoid outliers driving our results.

Figs. A1 and A2 show the share of tourism of GDP across countries and time, respectively. On average, international tourism contributes to around 5% of GDP but there is a wide range across countries. Tourism GDP has remained relatively stable over time for the median country, while the countries at the top 25% of the distribution have increased their reliance on tourism slightly over time until 2019. In 2020, due to the COVID-19 pandemic, tourism share of GDP dropped markedly.

It is more difficult, however, to obtain data on the GDP contribution of hotels. By combining the tourist arrival data with the hotel data, we make a back-of-the envelope calculation which shows that the median of hotel sector's contribution to GDP is around 2% across countries, but with a large interquartile range between close to zero and 6%. According to this calculation, for the median country around 2/3 of the tourism expenditure is going to hotels. As an alternative measure, the "Week at the Beach" index (Laframboise et al., 2014) shows that the cost of hotel on average accounts for 2/3 of the total cost for a typical 7-day stay at a beach destination which also includes meals, beverage and local transportation.

Table A1 displays the summary table for the tourism regression sample. On average, countries see an increase in tourist arrivals by 0.75%, which is consistent with the tourism increasing in importance over time. Fig. A3 shows indeed that tourist arrivals as a share of the population has increased over our sample period. The summary statistics on exchange rates show that on average, as expected, the impact of exchange rates is small, but with standard deviation of between 9% and 12%, depending on the exchange rate considered.

Fig. A3 shows hotel prices over time in U.S. dollars based on the Tripadvisor data. Hotel prices have stayed roughly stable in U.S. dollars. However, the standard deviation across countries is large. While the median price is at around 100 U.S. dollars across countries, the interquartile ranges from 120 to around 270 U.S. dollars. Fig. A4 also provides a distribution of hotel prices across hotels with different quality. The median hotel price for low quality hotels is less than 100 U.S. dollars, the medium around 120 U.S. dollar, and the high quality around 225 U.S. dollars.

3. Evidence of dominant currency pricing in hotel prices

Intuitively, the cost of accommodation is often one of the most important factors for tourists to consider when selecting international travel destinations from a group of countries that provide similar tourism products. Unlike the cost of airfare, which depends largely on global factors such as the oil price and profit margins of international airlines, the cost of accommodation depends largely on local factors such as local labor and utility costs, insurance expenses, and taxation. Which currency (or currencies) local hotels choose to invoice their services also matters, as the choice affects the actual cost for tourists via the exchange rate channel. If hotel prices are completely invoiced in destination-country currency (or PCP), one would not expect any correlation between the U.S. dollar exchange rate and hotel prices in local currency. Conversely, if hotel prices are invoiced in a foreign currency—say, the U.S. dollar—and are sticky, there should be a full pass-through from the fluctuations of the U.S. dollar exchange rate into domestic hotel prices.

In reality, the choice of invoicing currency at the country level can be a mix of the tourist destination (tourism exporting country) currency—PCP, one or more dominant currencies—DCP, and even the currency of the tourist origin (tourism importing country) country—the so-called local currency pricing (LCP). To quantify the elasticity of hotel prices with respect to exchange rate movements of different currency pairs, we estimate the following pass-through regression:

² Countries in currency unions (e.g. the Euro Area) are treated separately but exchange rate changes between countries are set to zero.

³ In collaborating with Tripadvisor, Laframboise et al. (2014) use the company's hotel price data to construct a "Week at the Beach" index which tracks the nominal cost of a one-week beach holiday in a tourism destination, including the average price of hotels with a three to four "bubble" rating together with more than 80 million crowdsourced data points on prices for meals, taxis, and beverages (water, coffee, and beer).

$$\Delta \text{HotelPrice}_{i,t} = \alpha + \alpha_i + \alpha_t + \beta_1 \Delta \text{FX}_{i,t} + v_{i,t}$$

where $\Delta \text{HotelPrice}_{i,t}$ is the change in the log hotel price of hotels in country i in year t in destination-country currency, $\Delta \text{FX}_{i,t}$ is the change in the log of destination-country currency units relative to the U.S. dollar (a positive value indicates a depreciation against the dollar), and α_i and α_t represent country and year fixed effects, respectively. If all hotels were priced in the domestic currency (PCP) and sticky, a depreciation of the domestic currency against the dollar is not expected to affect hotel prices in domestic currency ($\beta_1 = 0$). Conversely, if hotel prices were invoiced in the U.S. dollar (DCP), a domestic depreciation against the dollar would lead to full pass-through of the exchange rate movement to domestic hotel prices ($\beta_1 = 1$).

The results of the above regression are presented in Table 1. Column (1) shows the results for all hotels without any fixed effects. The estimated coefficient of β_1 is positive (0.45) and statistically significant, indicating that a domestic depreciation against the dollar indeed increases hotel prices in domestic currency. The fact that the coefficient is statistically different from zero suggests that hotel prices are partly invoiced in the U.S. dollar and sticky. One caveat could be that hotel prices increase in general in times when the U.S. dollar appreciates. To control for this possibility, we include year fixed effects in the specification. One other concern could be that country-specific factors (e.g., the growth rate of the country over our time period) are correlated with the movements in the U.S. dollar and could lead to a spurious correlation between U.S. dollar movements and hotel prices. To address this concern, we also estimate a within-country regression by including country fixed effects. All results hold and remain virtually unchanged when country, year, and both fixed effects are considered (columns 2 through 4, respectively).

In columns (5) through (7), the sample is split according to the Tripadvisor bubble ratings of the hotels: high-quality in column (5), medium-quality hotels in column (6), and low-quality hotels in column (7). The pass-through coefficient is the smallest for the medium-quality hotels, followed by that of the high-quality and low-quality hotels. Because a larger coefficient can be interpreted as representing stronger evidence of DCP, the results suggest that medium-quality hotels are most likely to be priced in domestic currency, while DCP seems to be most common among low-quality hotels. The standard error in column (7), however, is relatively large, and the coefficient therefore is not statistically different from that in column (5) or (6). In all columns, the hypothesis of $\beta_1 = 1$, or full DCP, can also be rejected.

In Fig. 1, we present a binscatter plot between the bilateral percentage depreciation of destination-country currency against the dollar and the percentage change in hotel prices in domestic currency. The positive relationship indicates that hotel prices in the domestic currency increase when the domestic currency depreciates, which again provides strong evidence against full PCP in the hotel sector. On the other hand, the linearly fitted line of the positive relationship is significantly flatter than a 45 degree, indicating that the sample does not imply a full DCP either.

Figs. 2 through 4 demonstrate the heterogeneity in the elasticity of hotel prices with respect to exchange rate movements at the country level. We estimate a time-series regression for each country in the sample separately and regress the change in the log hotel price on the change in the log exchange rate relative to the U.S. dollar without country fixed effects. The estimated β_1 coefficient thus reflects the country-specific elasticity, which ranges from negative 0.5 to positive 3.2 (Fig. 2). As shown in Fig. 3, while there is large heterogeneity across countries, the most frequent observations center around β_1 of 1, or full DCP, and the second most frequent observations center around β_1 of 0. This bimodal distribution suggests that, at the country level, there can be a strong concentration around full DCP or full PCP depending on country-specific circumstances. The European countries, for instance, would have little incentive to invoice their hotels in a currency other than the euro, given the large share of tourists within the euro area. This result may also explain why the elasticity does not vary significantly across quality groups in the above analysis.

Finally, we sort the country-specific elasticity by the size of their gross domestic product (GDP) in Fig. 4. Although the overall sample is skewed toward small tourism-dependent economies, larger countries (especially those in the euro area) tend to have a smaller coefficient, an indication of weaker DCP in the prices of their hotel sector.

4. Evidence of exchange rate elasticities in tourist arrivals

In this section, we use the bilateral tourist arrival data from the UNWTO to quantify the elasticity of the volume of tourist arrivals with respect to exchange rate movements. The data comprises 181 destination and 200 origin countries from 1995 until 2019. Using bilateral tourist arrival data allows us to control for time-varying destination-specific shocks in the regression, so that we can exploit the heterogeneity in exchange rate movements among different currency pairs that may affect the volume of tourist arrivals and shed light on the effect of DCP on the quantity of international tourism.

To quantify the effect of exchange rate movements on tourist arrivals, we estimate the following regression:

$$\Delta \text{Arrival}_{i,j,t} = \alpha + \alpha_t + \alpha_{ij} + \beta_1 \Delta \text{FX}_{i,j,t} + \beta_2 \Delta \text{USD}_{j,t} + \beta_3 \Delta \text{DFX}_{i,j,t} + v_{i,j,t}$$

where $\Delta \text{Arrival}_{i,j,t}$ is the yearly difference in the log number of tourists arriving at destination country i from origin country j between year $t-1$ and t . α_t are year fixed effects that capture all time-variant global shocks, such as the movement of the U.S. dollar against all countries. α_{ij} are origin-country times destination-country fixed effects that control for any time-invariant factors of the origin–destination country pair that is typically included in gravity model analysis, such as the distance between the two countries. $\Delta \text{FX}_{i,j,t}$ is the yearly difference between the log of the destination-country currency relative to the origin-country currency (e.g., the Fijian dollar relative to the euro for tourists arriving from Germany to Fiji), where a positive value reflects a depreciation of the origin country relative to the destination country. $\Delta \text{USD}_{j,t}$ captures the movement

Table 1
Hotel Price Pass-Through Regression.

	Dependent Variable: Δ Local Hotel Prices						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
% Depreciation against USD	0.452*** (0.081)	0.460*** (0.080)	0.473*** (0.090)	0.483*** (0.090)	0.455*** (0.103)	0.308*** (0.090)	0.504* (0.252)
N	3,543	3,543	3,543	3,543	3,403	3,447	2,063
R2	0.017	0.021	0.080	0.084	0.059	0.042	0.054
Hotels	All	All	All	All	High Quality	Medium Quality	Low Quality
Country FE	–	Y	–	Y	Y	Y	Y
Time FE	–	–	Y	Y	Y	Y	Y
P-Value: $b = 1$	0.000	0.000	0.000	0.000	0.000	0.000	0.058

Note: This table shows the results from a hotel price pass-through regression. The dependent variable is the change in the log average hotel price in local currency of a country i where the hotels are based between year t and $t-1$. The independent variable is the change between the log exchange rate of country i relative to the U.S. dollar (USD). Column (1) does not include fixed effects. Column (2) includes country i fixed effects. Column (3) includes year t fixed effects. Column (4) includes country i and year t fixed effects. In column (5), the dependent variable is the change in the log hotel price for only high-quality hotels. In column (6), the dependent variable is the change in the log hotel price for only medium-quality hotels. In column (7), the dependent variable is the change in the log hotel price for only low-quality hotels.

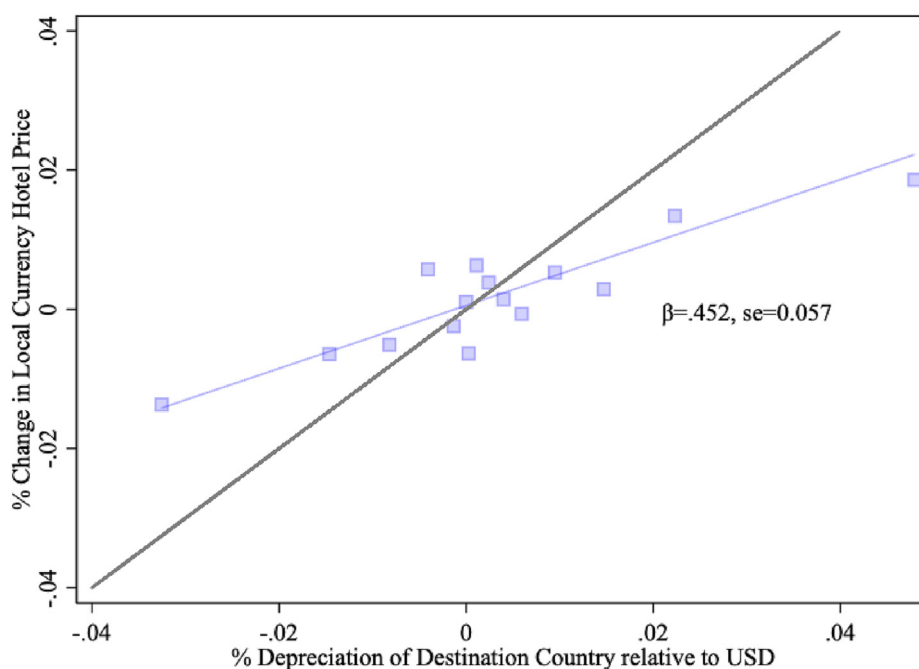


Fig. 1. Hotel Price Sensitivity. Note: This graph shows a binscatter plot between the percent depreciation of the destination country relative to the U.S. dollar (on the horizontal axis) and the percent change in local currency hotel prices (on the vertical axis).

of the origin-country currency relative to the U.S. dollar (e.g., the euro relative to the U.S. dollar for German tourists visiting Fiji). $\Delta DFX_{i,j,t}$ captures the movement in the origin-country currency relative to the country-specific dominant currency (CSDC) of destination country i . The CSDC is defined as the currency of the country where the largest share of tourists originates from for a given destination country. In the case of Fiji, the CSDC (and the dominant currency country) is the Australian dollar (and Australia). Standard errors are clustered at the country-pair level.

This regression equation is similar to that in (Gopinath et al., 2020a), which regresses import quantities on the exchange rate movement of the exporter relative to the U.S. dollar and the bilateral exchange rate.⁴ However, our regression specification has an additional term $\Delta DFX_{j,t}$ given that there may exist multiple dominant currencies in the global tourism market because tourism products can be region specific, while the U.S. dollar is perceived as the only dominant currency in international trade of goods.

⁴ Gopinath, Li, and Meleshchuk (2020b) use the European Union data to analyze exchange rate elasticities for tourism.



Fig. 2. Country-Specific Elasticity of Hotel Prices (in Local Currency) to Exchange Rate. Note: This graph shows the estimated coefficient of a time-series regression for each country of percent change in local currency hotel prices on the percent depreciation of the destination country relative to the U.S. dollar.

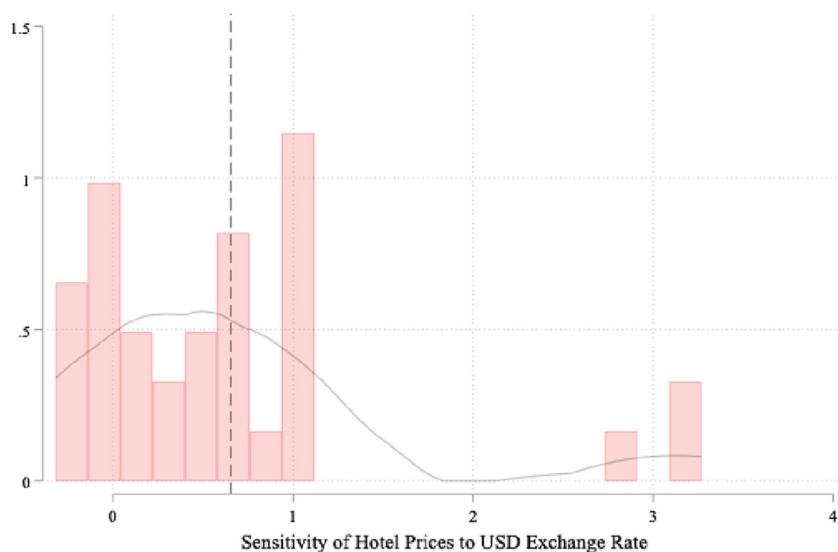


Fig. 3. Country-Specific Elasticity of Hotel Prices (in Local Currency) to Exchange Rate. Note: This graph shows a histogram of the coefficient of a time-series regression for each country of percent change in local currency hotel prices on the percent depreciation of the destination country relative to the U.S. dollar.

We first present the result for each currency pair separately in Table 2. Column (1) shows the results for the bilateral exchange rate movements, $\Delta FX_{i,j,t}$. A bilateral depreciation of the origin country relative to the destination country is associated with a decline in tourist arrivals from the given origin country to the given destination country, consistent with a downward-sloping demand curve and PCP, as one would expect under the standard Mundell-Fleming framework. (If all inter-

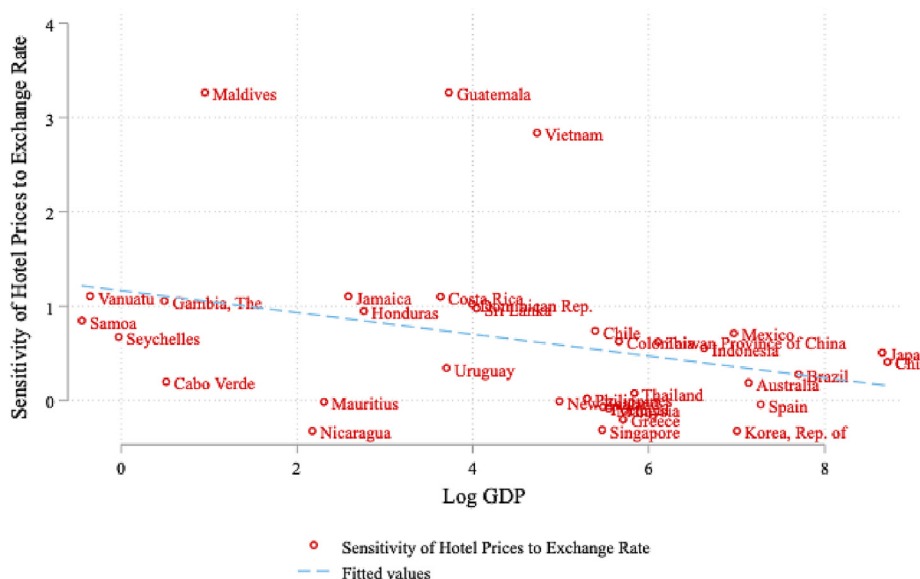


Fig. 4. Sensitivity of Hotel Prices and Country Size. Note: This graph shows a scatterplot between the the coefficient of a time-series regression for each country of percent change in local currency hotel prices on the percent depreciation of the destination country relative to the U.S. dollar (on the vertical axis) and log gross domestic product (GDP) of the country (on the horizontal axis).

Table 2
Tourist Arrival Regressions.

	Dependent Variable: Δ Arrivals			
	(1)	(2)	(3)	(4)
Δ Bilateral Exchange Rate	-0.109*** (0.009)			-0.073*** (0.014)
Δ USD vs. Origin Country		-0.187*** (0.014)		-0.109*** (0.021)
Δ Dominant Currency vs. Origin Country			-0.096*** (0.010)	
N	225,086	233,108	240,705	220,888
R ²	0.063	0.061	0.062	0.064
Time FE	Y	Y	Y	Y
Country Pair FE	Y	Y	Y	Y

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. Time (year) and country-pair fixed effects are included in all columns.

national travel were priced in destination-country currency, a bilateral depreciation in origin-country currency would increase prices for travelers in their own currency.) Quantitatively, our results suggest that a 10 percent depreciation of origin-country currency relative to destination-country currency is associated with a 1.1 percent decline in the volume of tourists.

Column (2) shows the result for the U.S. dollar exchange rate, Δ USD $_{j,t}$. The estimated elasticity coefficient is also negative, indicating that when the origin-country currency depreciates relative to the U.S. dollar, travel from origin country j to destination country i declines, even when Δ USD $_{j,t}$ is not directly related to destination country i . This result can be interpreted as an indication of DCP. Namely, if hotel prices are set in the U.S. dollar in destination-country i in year t and are sticky between t and $t-1$, a depreciation of the origin-country currency relative to the U.S. dollar increases hotel prices for tourists and therefore reduces demand, or tourism flows, from origin-country j to destination-country i . Quantitatively, a 10 percent depreciation of the origin-country currency relative to the U.S. dollar reduces travel flows by 1.9 percent.

Column (3) shows the results for the CSDC, Δ DFX $_{j,t}$. The dominant currency is defined as the change in the log exchange rate between the dominant currency of country i and origin country i (e.g., the euro against the Australian dollar for German tourists visiting Fiji). The coefficient is also negative but smaller in absolute values relative to that in column (2) for the U.S.

dollar exchange rate. A 10 percent depreciation of the origin-country currency relative to the CSDC reduces travel by around 1 percent.

The three exchange rates are naturally highly correlated. In fact, for many tourism destinations, especially those in the Caribbean, the United States is their CSDC country, in which case $\Delta DF_{i,j,t}$ and $\Delta USD_{j,t}$ are perfectly collinear. We therefore include all three exchange rates in the same regression. The results are shown in column (4) of Table 2 and graphically in Fig. 5, which indicate that the effect of CSDC completely vanishes once one controls for the U.S. dollar exchange rate. Both the U.S. dollar exchange rate and the bilateral exchange rate between origin country and destination country remain statistically and around equally important.⁵

Although our regressions control for both time and country fixed effects that would account for any time-invariant country characteristics and global shocks that could affect overall demand for international travel, the regressions do not control for country-specific time-varying shocks, such as a negative demand shock, as they could be collinear with the movements in the exchange rates. We therefore estimate the following equation:

$$\Delta \text{Arrival}_{i,j,t} = \alpha + \alpha_{i,t} + \alpha_{ij} + \beta_1 \Delta e + v_{i,j,t}$$

where Δe is either ΔFX , ΔUSD or ΔDF , $\alpha_{i,t}$ are destination country times year fixed effects that can capture all time-variant and time-invariant destination-country-specific shocks. As these regressions include destination country times time fixed effects, the average change of tourist arrivals to a destination country across all origin countries and other destination-specific time-varying factors, such as the change in the price level, are controlled for. Therefore, potential confounding factors that would lead to a depreciation of the currency and reduce travel demand, such as social-political uncertainty or inflation, will not be driving the results.

This regression specification cannot be estimated as a horse-race regression between the bilateral exchange rate and the U.S. dollar exchange rate, as $\Delta FX_{i,j,t}$ and $\Delta USD_{j,t}$ would be perfectly collinear after including destination-country times year fixed effects. The movement between the origin-country currency against the U.S. dollar is a linear combination of the bilateral movement between destination country and origin country and the U.S. dollar movement against the destination country (which is absorbed by destination-country times year fixed effect). The results corroborate our previous finding that the U.S. dollar exchange rate (between the origin-country currency and the U.S. dollar) remains important, especially when compared with the bilateral exchange rate and the CSDC exchange rate (Table 3).⁶

We next analyze the dynamic effects of exchange rate changes on tourist arrivals. We estimate local projections by regressing the cumulative tourist flows over multiple year horizon. Figure A6 shows the results for the bilateral exchange rate and the U.S. dollar. By definition, for year zero, the effect is the same as for our baseline specification, in which both the depreciation of the U.S. dollar and of the domestic currency lead to increased arrivals of tourists. The effect strengthens for the next year for both exchange rates and remains slightly stronger for the U.S. dollar but remains statistically significant for both. After the first year, the effect levels off, but does not seem to revert back. This leads us to conclude that a one off depreciation of the exchange rate can persistently increase tourist arrivals.

5. Dominant currency financing

Having presented the evidence of DCP in both prices and quantities of international tourism, we investigate in this section the complementarities between pricing (DCP) and financing (DCF), which may help explain the important role of the U.S. dollar in tourism regardless of the origin or destination countries of international tourists, similar to the findings of Gopinath and Stein (2020) and Adler et al. (2020). The fact that the U.S. dollar exchange rate tends to be more important than the CSDC exchange rate suggests that factors unique to the U.S. dollar may be at play. One potential hypothesis put forward in the literature is that U.S. dollar pricing and financing complement each other. For example, Casas et al. (2020) show that exporting in the U.S. dollar provides a natural hedge against a depreciation for firms borrowing in the U.S. dollar. Given the uniquely important role of the dollar in international finance—especially for firms in emerging market and developing countries, including those in the tourism industry—it is conceivable that the need to service the dollar debt gives tourism exporters incentives to price their products in the dollar.

We test whether tourism destinations that borrow more in the U.S. dollar are more sensitive to movements of the dollar exchange rate. We start by investigating the pricing response of hotels in response to the U.S. dollar exchange rate as a function of U.S. dollar borrowing, followed by quantity regressions where we investigate whether tourism arrivals react differently to the U.S. dollar exchange rate when the destination countries have more U.S. dollar debt. Unfortunately, we do not have data on U.S. dollar borrowing of hotels. The best approximation we have is the data on U.S. dollar borrowing of non-financial corporates as compiled by Adler et al. (2020). While we are aware that U.S. dollar borrowing of nonfinancial corporates is not a perfect proxy of U.S. dollar borrowing by hotels, the imperfect proxy is unlikely to bias our results in a systematic way and instead, if anything, attenuates our effects.

⁵ Note that the number of observation differs due to the trimming of the exchange rate variables and the inclusion of fixed effects that lead to singleton observations that are not included in the regression specification.

⁶ Note that the number of observations differ relative to other tables due to the inclusion of country-time fixed effects leading to more singleton observations, which are not included in the regression.

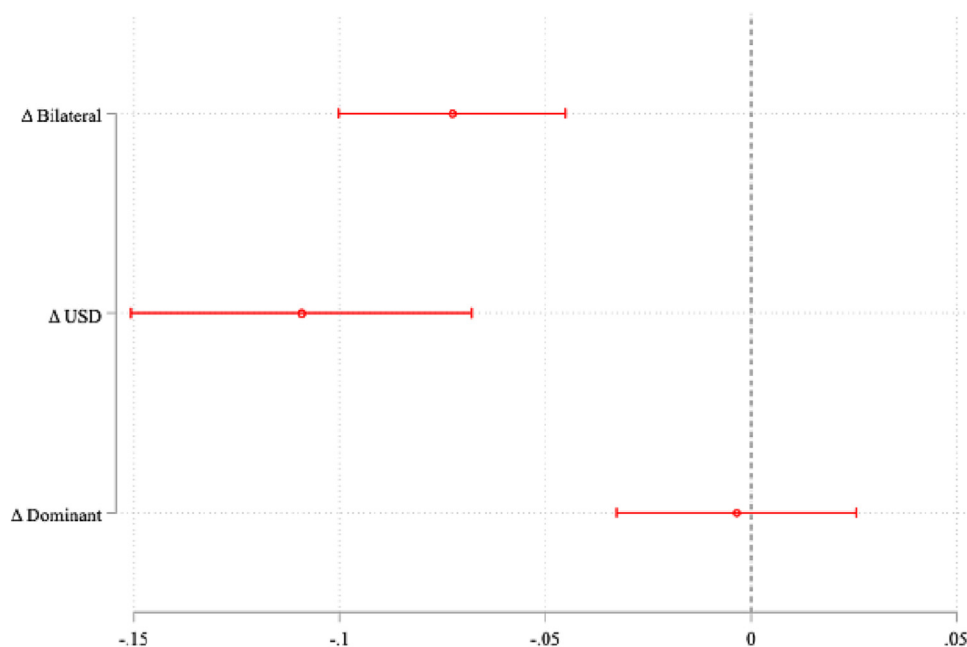


Fig. 5. Exchange Rate Elasticity of Tourism Arrivals. Note: This graph shows the the estimated effects of the percent bilateral depreciation of the origin country relative to the destination country, the percent depreciation of the origin country relative to the U.S. dollar, and the percent depreciation of the origin country relative to the (destination) country-specific dominant currency on tourist arrivals from origin country to destination country.

Table 3
Tourist Arrival Regressions.

	Dependent Variable: Δ Tourist Arrivals		
	(1)	(2)	(3)
Δ Bilateral Exchange Rate	-0.168*** (0.012)		
Δ USD vs. Origin Country		-0.191*** (0.013)	
Δ Dominant Currency vs. Origin Country			-0.170*** (0.013)
N	225,069	233,092	240,672
R2	0.152	0.152	0.151
Country*Time FE	Y	Y	Y
Country Pair FE	Y	Y	Y

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. $\Delta Bilateral\ Exchange\ Rate$ is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. $\Delta USD\ vs.\ Origin\ Country$ is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. $\Delta Dominant\ Currency\ vs.\ Origin\ Country$ is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. Destination-country*Time (year) and country-pair fixed effects are included in all columns.

5.1. Hotel prices

We estimate a similar equation as in section II but introduce an additional interaction term:

$$\Delta HotelPrice_{i,t} = \alpha + \alpha_i + \alpha_t + \beta_1 \Delta FX_{i,t} + \beta_2 \Delta FX_{i,t} * USD\ Borrowing_{i,t} + \beta_3 USD\ Borrowing_{i,t} + v_{i,t}$$

where $USD\ Borrowing_i$ is the demeaned share of dollar borrowing by country i in each year, as described in Adler et al. (2020). U.S. dollar borrowing is defined in various ways, as explained below. Our main coefficient of interest is β_2 , which reflects the additional effect of the exchange rate movement when countries have more U.S. dollar borrowing.

If the complementarity between U.S. dollar financing and pricing exists in the tourism industry, one would expect to see more dollar invoicing of hotel prices in countries with higher U.S. dollar debt. When a country's exchange rate depreciates, the servicing cost of its U.S. dollar debt in local currency increases. If the debtor (such as a hotel) does not have offsetting revenues in the dollar, the depreciation will lead to a negative net worth impact. However, if the revenue of the debtor is

also in the dollar (by invoicing hotel services in the dollar), the higher revenues in local currency can provide a hedge against the increase in debt repayments in local currency terms (Casas et al., 2020).

As U.S. dollar pricing would be reflected in a positive coefficient for $\Delta FX_{i,t}$, we would expect β_2 to be positive as well, as the effect would be stronger in countries with more U.S. dollar borrowing. The results are shown in Table 4. Column (1) shows the results for all hotels, while columns (2) through (4) split by hotel rating groups. The coefficient of interest, β_2 , is indeed positive and statistically significant in columns (1) through (3), suggesting strong complementarities between DCP and financing for higher-level hotels. For lower-rated hotels, the interaction between the share of U.S. dollar borrowing and the exchange rate is insignificant, potentially due to their inability to borrow in U.S. dollars.

Quantitatively, as we demean the U.S. dollar borrowing we can interpret the coefficient on the exchange rates themselves as the effect of a country with the average share of U.S. dollar borrowing. The effect of a 10 percentage point higher share of U.S. dollar borrowing around one standard deviation is one tenth of the interaction term with the total effect the sum of those. For instance, on average, the price elasticity with respect to exchange is 0.36 for a country with the average share of U.S. dollar borrowing, while the elasticity increases to an elasticity of 0.55 ($0.36 + 0.1 \cdot 1.908$).

The results can also be demonstrated in a binscatter in Fig. 6. As for the pooled country sample, we plot a binscatter of the percent change in local currency hotel prices on the percent depreciation of destination country relative to the U.S. dollar, and we split the sample between destination countries that borrow heavily in U.S. dollar and those that do not. The high dollar-borrowing countries are displayed in gray diamonds, and the low dollar-borrowing countries are shown as red circles. The positive correlation between the depreciation of the exchange rate and the rise in domestic currency hotel prices is entirely driven by firms that borrow heavily in the dollar. When we compare the slope of countries that do not borrow much in the dollar, the correlation is much stronger for high dollar-borrowing countries than for low dollar-borrowing ones.

5.2. Tourist arrivals

We now test whether complementarities between DCF and DCP also affect the tourist arrivals. We estimate the following regression:

$$\Delta Arrival_{i,j,t} = \alpha + \alpha_t + \alpha_{ij} + \beta_1 \Delta FX_{i,j,t} + \beta_2 \Delta USD_{i,j,t} + \beta_3 \Delta DFX_{i,j,t} + \beta_4 \Delta FX_{i,j,t} * USD \text{ Borrowing}_{i,t} + \beta_5 \Delta USD_{i,j,t} * USD \text{ Borrowing}_{i,t} + \beta_6 \Delta DFX_{i,j,t} * USD \text{ Borrowing}_{i,t} + \beta_7 USD \text{ Borrowing}_{i,t} + v_{i,j,t}$$

The coefficients of interest are β_4 through β_6 . They indicate to which exchange rate movements countries with more foreign currency borrowing will be more sensitive. In all regressions, as shown in Table 5, we can observe that the U.S. dollar exchange rate becomes more important for countries that borrow more in the dollar, mirroring the evidence for hotel prices in the previous section. In contrast, the bilateral exchange rate becomes less important for these countries, whereas the dominant exchange rate effect is unchanged as a function of U.S. dollar borrowing.

Fig. 7 displays the results graphically by calculating the effect of the bilateral and the U.S. dollar exchange rate on tourist arrivals for high and low U.S. dollar borrowing countries. The red bars show the differential effect of the U.S. dollar exchange rate on tourism arrivals, differentiating between high dollar-borrowing countries (shaded) and low dollar-borrowing countries (solid). The effect of the U.S. dollar exchange rate is significantly stronger and almost twice as large quantitatively in high dollar-borrowing countries. The blue bars show the effects of the bilateral exchange rate. The opposite pattern can be seen here. The effect of the bilateral exchange rate is significantly stronger when countries do not borrow heavily in the U.S. dollar (solid bar), with a much more muted effect for countries that borrow heavily in the dollar.

6. Country-Specific dominant currencies

So far, we have established that the U.S. dollar plays a special role in the pricing of tourism, likely due to its importance as a financing currency. In contrast, the currency of the country where most tourists originate from, or CSDC, on average, plays a smaller role when the currency is not the U.S. dollar. Fig. 8 displays the CSDC for selected countries. For instance, the United States is the dominant currency country for Canada and Mexico, while the CSDC for the United States is the Canadian dollar. In many African countries, the dominant currency country is often France or Germany, and the euro is therefore the CSDC. China accounts for the largest share of tourists for many Asia and Pacific countries, including Australia, while Australia is the dominant currency country for New Zealand and most Pacific Island countries.

Intuitively, when there is a high degree of concentration of foreign tourists, hotels may have incentives to set prices in the CSDC to stabilize the price for their largest markets. Fig. 9 shows the share of tourists arriving from the CSDC country for each destination country. Taking together Figs. 8 and 9, we can see that, for example, the largest share of tourists to New Zealand is from Australia, and this share is relatively large (40 percent of New Zealand's total tourist arrivals), indicating that New Zealand hotels may have a stronger incentive to invoice their rates in the Australian dollar. A strengthening of the Australian dollar relative to other currencies would therefore increase the costs for tourists from other countries to New Zealand. In contrast, while the largest share of tourists to Australia is from China, this share is relatively small (less than 20 percent), hence the incentives for Australian hotels to price their rates in the renminbi may be low.

To test whether the CSDC indeed becomes more important with the market share of the dominant currency country, we estimate the following regression:

Table 4

Hotel Price Pass-Through Regression and U.S. Dollar Borrowing.

	Dependent Variable: Δ Local Hotel Prices			
	(1)	(2)	(3)	(4)
% Depreciation against USD	0.363*** (0.076)	0.347*** (0.078)	0.266** (0.099)	0.502*** (0.157)
% Depreciation against USD*USD	1.908*** (0.660)	2.078*** (0.636)	1.587** (0.614)	1.895 (3.969)
N	1,499	1,456	1,499	1,251
R ²	0.139	0.130	0.088	0.077
Hotels	All	High Quality	Medium Quality	Low Quality
Country FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y

Note: This table shows the results from a hotel price pass-through regression. The dependent variable is the change in the log average hotel price in local currency in local currency of a country i where the hotels are based between year t and $t-1$. % *percent Depreciation against USD* is the change between the log exchange rate of country i relative to the U.S. dollar (USD). *USD* is the demeaned share of USD Borrowing of companies in country i , as described in [Adler et al. \(2020\)](#). The dependent variable in column (1) is the log change in the local currency hotel price across all hotels, in column (2) for only high-quality, column (3) only for medium quality, column (4) for low quality hotels.

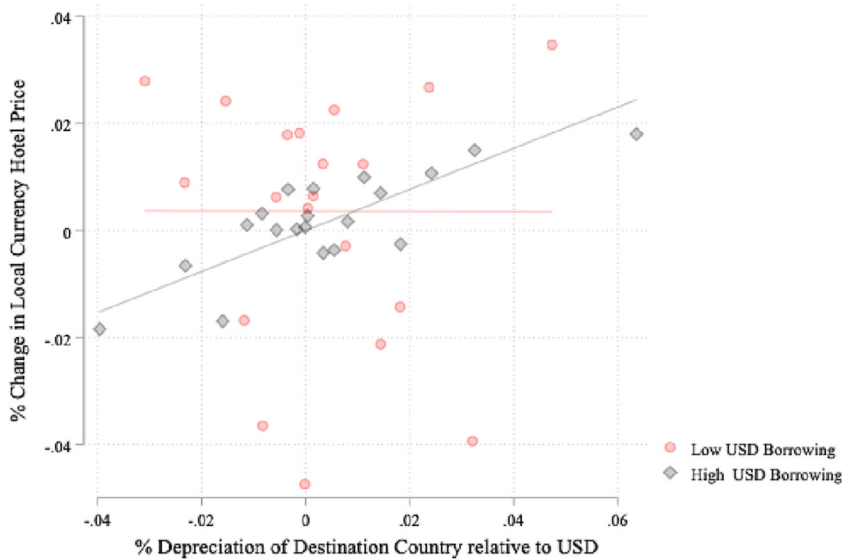


Fig. 6. Sensitivity of Hotel Prices and U.S. Dollar Borrowing. Note: This graph shows a binscatter plot between the percent depreciation of the destination country relative to the U.S. dollar (USD) (on the horizontal axis) and the percent change in local currency hotel prices (on the vertical axis) split by countries with a high (black diamonds) and low (red dots) USD borrowing. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\Delta Arrival_{i,j,t} = \alpha + \alpha_t + \alpha_{ij} + \beta_1 \Delta FX_{i,j,t} + \beta_2 \Delta USD_{i,j,t} + \beta_3 \Delta DFX_{i,j,t} + \beta_4 \Delta FX_{i,j,t} * Concentration_{i,t} + \beta_5 \Delta USD_{i,j,t} * Concentration_{i,t} + \beta_6 \Delta DFX_{i,j,t} * Concentration_{i,t} + \beta_7 Concentration_{i,t} + v_{i,j,t}$$

where $Concentration_{i,i}$ is the share of tourists arriving from the CSDC country for each destination country i . Column (1) of [Table 6](#) shows the results. Consistent with the intuition, the importance of the CSDC increases with the degree of concentration of tourists for destination countries, while the importance of the bilateral exchange rate decreases. In column (2), we replace the share of tourists arriving from the CSDC country with a dummy that is one if the share of tourists arriving from the dominant currency country is high. The results are qualitatively the same and are illustrated in [Fig. 11](#). The solid bars reflect the role of the three different exchange rates when the concentration of tourist arrivals is high. The U.S. dollar and the exchange rate of the dominant country play a large and quantitatively similar role, while the bilateral exchange rate becomes irrelevant. This result further highlights that, when tourist arrivals are highly concentrated, tourism destination countries can do little to improve competitiveness via a domestic exchange rate depreciation. On the other hand, in countries where tourist arrivals are not highly concentrated, the bilateral exchange rate can still act as an effective shock absorber. Columns (3) and (4) show that the results hold when we exclude the United States as a dominant country, and column (5) shows that the results are robust when including controls.

Column (6) replaces the concentration measure with a measure of tourist reliance, defined as the annual tourist arrivals as a share of the destination-country population. The tourist reliance variable is shown in a map in [Fig. 10](#). Similar to the

Table 5
Tourist Arrival Regression: The Role of Dominant Currency Financing.

	Dependent Variable: Δ Tourist Arrivals			
	(1)	(2)	(3)	(4)
Δ Bilateral Exchange Rate	-0.099*** (0.023)	-0.170 (0.141)	-0.094*** (0.021)	-0.004 (0.138)
Δ Bilateral Exchange Rate*USD	0.324*** (0.093)	0.872*** (0.145)	0.499*** (0.094)	0.677*** (0.123)
Δ USD vs. Origin Country	-0.093*** (0.028)	0.167 (0.138)	-0.111*** (0.028)	0.137 (0.140)
Δ USD vs. Origin Country*USD	-0.215** (0.101)	-0.297** (0.140)	-0.428*** (0.083)	-0.194* (0.113)
Δ Dominant Currency vs. Origin Country	-0.043** (0.021)	-0.439*** (0.121)	-0.032 (0.021)	-0.551*** (0.125)
Δ Dominant Currency vs. Origin Country*USD	-0.249*** (0.091)	-0.717*** (0.130)	-0.162* (0.084)	-0.536*** (0.121)
N	87,602	71,239	87,602	71,239
R ²	0.067	0.081	0.068	0.081
Time FE	Y	Y	Y	Y
Country Pair FE	Y	Y	Y	Y
Measure	Share Liabilities	Share Liabilities	Share Debt	Share Debt
Controls	-	Y	-	Y

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. USD indicates various measure of U. S. dollar borrowing. Columns (1) and (2) define USD as the share of external liabilities in U.S. dollars. Columns (3) and (4) define USD as the share of external debt liabilities in U.S. dollars. Controls include the interaction between GDP growth, financial integration, GDP per capita, trade integration, and the size of the economy with the exchange rate changes. Time (year) and country-pair fixed effects are included in all columns.

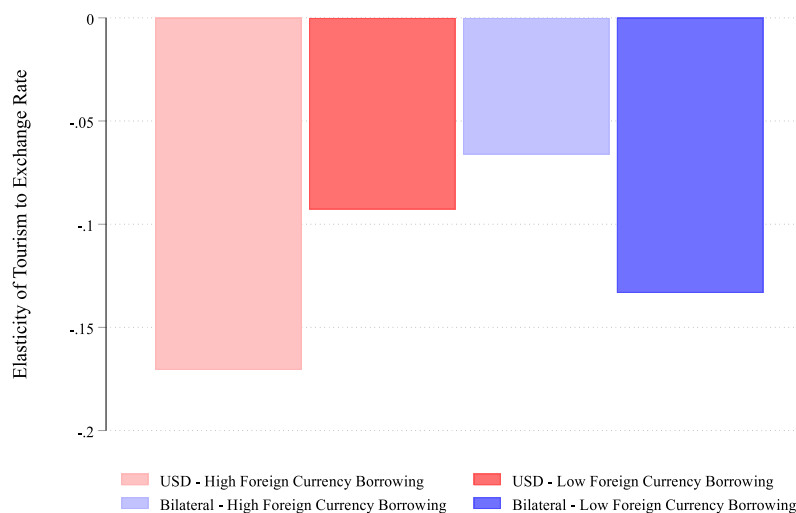


Fig. 7. The Exchange Rate Elasticity of Tourist Arrivals and U.S. Dollar Borrowing. Note: This graph shows the estimated effects of the percent bilateral depreciation of the origin country relative to the destination country (blue) and percent depreciation of the origin country relative to the U.S. dollar (USD) (red) for the destination country with high (above median) in shaded colors and low (below median) U.S. dollar borrowing in solid colors on tourist arrivals from origin country to destination country. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

concentration measures, countries with higher tourist reliance (or a larger size of tourists relative to the local population), including many tourism-dependent economies in the Caribbean and the Pacific, are more sensitive to changes in the exchange rate of the CSDC.

In Table 7, we split the sample by dominant countries to assess whether there exist currencies other than the U.S. dollar that play an important role in their respective segments of the international tourism markets. We find that in countries where the dominant currency country is the United Kingdom and Russia, the pound and the ruble do play a statistically significant role, even for tourists originating from other countries (Table 7). However, the sample size is relatively small for the pound and the ruble markets, while currencies such as the Australian dollar, the euro, and the renminbi do not exhibit

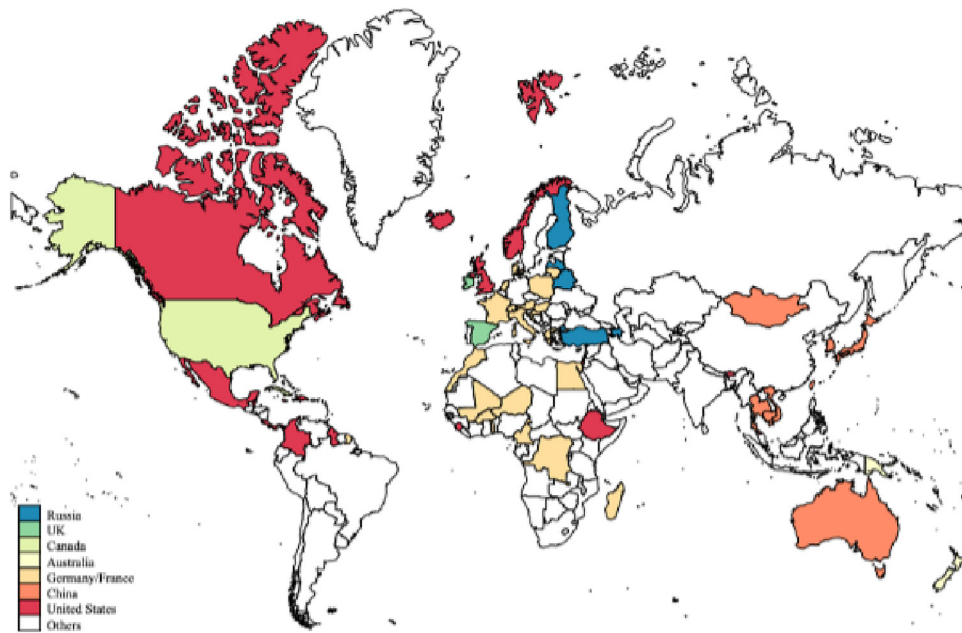


Fig. 8. Dominant Currency Country. Note: This map indicates the country where the largest share of tourists are originating from.

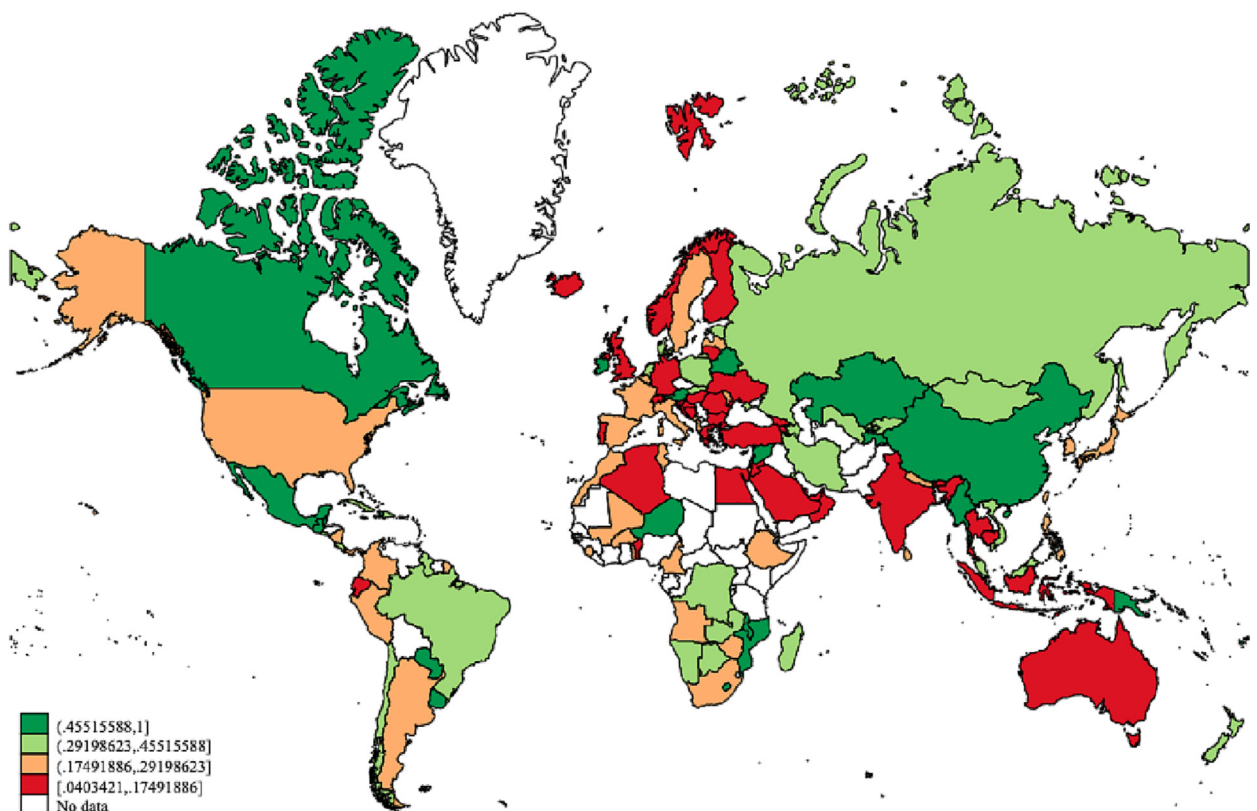


Fig. 9. Concentration of Tourist Arrivals. Note: This map indicates the share of tourists arriving from the country where the largest number of tourists are arriving from.

Table 6

Tourist Arrival Regression: The Role of Concentration.

	Dependent Variable: Δ Tourist Arrivals					
	(1)	(2)	(3)	(4)	(5)	(6)
Concentration	−0.087*** (0.014)	−0.037*** (0.006)	−0.030*** (0.006)	−0.030*** (0.006)	−0.037*** (0.008)	0.006*** (0.001)
Δ Bilateral Exchange Rate	−0.171*** (0.026)	−0.084*** (0.015)	−0.059*** (0.014)	−0.059*** (0.014)	−0.365*** (0.067)	−0.036** (0.015)
Δ Bilateral Exchange Rate * Concentration	0.322*** (0.078)	0.106*** (0.041)	0.029(0.037)	0.029(0.037)	0.128*** (0.047)	−0.003(0.013)
Δ USD vs. Origin Country	−0.048(0.034)	−0.115*** (0.021)	−0.117*** (0.019)	−0.117*** (0.019)	0.254*** (0.076)	−0.141*** (0.021)
Δ USD vs. Origin Country * Concentration	−0.194** (0.094)	0.022(0.049)	−0.090** (0.042)	−0.090** (0.042)	−0.185*** (0.052)	−0.006(0.014)
Δ Dominant Currency vs. Origin Country	0.080*** (0.030)	0.032** (0.015)			−0.053(0.063)	
N	220,639	220,639	219,778	219,778	154,471	211,282
R ²	0.065	0.065	0.063	0.063	0.088	0.064
Time FE	Y	Y	Y	Y	Y	Y
Country Pair FE	Y	Y	Y	Y	Y	Y
Specification	Share Top	High Share	Share Top	High Share	High Share	Tourist Reliance
Origin Sample	All	All	Excl. US	Excl. US	All:Controls	All

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. Concentration indicates various measure of concentration. In columns (1) and (3), it is defined as the share of tourists arriving from the top origin country. In columns (2), (4), and (5), it is defined as a dummy that is one if the share of tourists arriving from the top origin country is above the median. In column (6), it is defined as the share of tourist arrivals per year over total population. Columns (3) and (4) exclude the United States as a origin country. Column (5) includes controls. Time (year) and country-pair fixed effects are included in all columns.

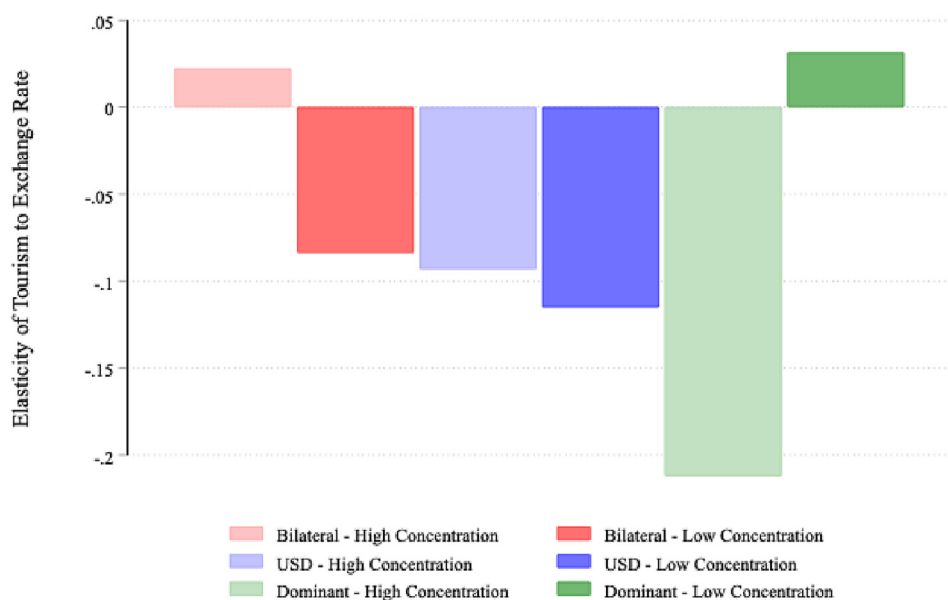


Fig. 11. The Exchange Rate Elasticity of Tourist Arrivals and Concentration. Note: This graph shows the estimated effects of the percent bilateral depreciation of the origin country relative to the destination country (red), percent depreciation of the origin country relative to the U.S. dollar (blue) for destination country and the percent depreciation of the origin country relative to the (destination) country dominant currency (green), with high (above median) in shaded colors and low (below median) concentration (share of tourists coming from the top country) in solid colors on tourist arrivals from origin country to destination country. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

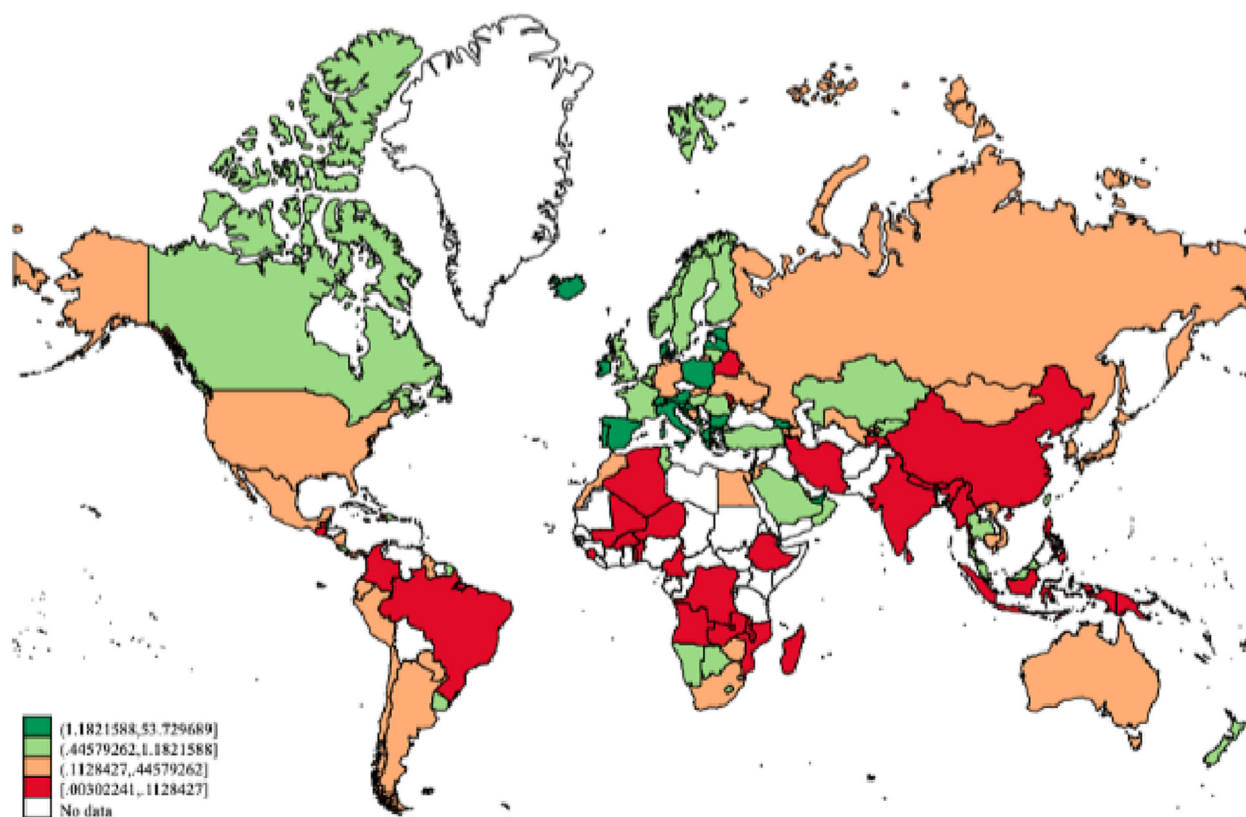


Fig. 10. Tourism Dependence. Note: This map indicates the share of annual tourist arrivals as a share of the local population.

Table 7

Tourist Arrival Regression, by Dominant Origin Country.

	Dependent Variable: Δ Tourist Arrivals						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ Bilateral Exchange Rate	-0.022 (0.154)	-0.247** (0.116)	-0.303*** (0.025)	0.033 (0.020)	-0.010 (0.063)	-0.012 (0.074)	0.269*** (0.043)
Δ USD vs. Origin Country	0.692 (0.683)	-0.014 (0.172)	0.124** (0.057)	-0.195*** (0.029)	0.313** (0.127)	0.001 (0.131)	-0.502*** (0.055)
Δ Dominant Currency vs. Origin Country	-0.694 (0.638)	-0.025 (0.115)	-0.048 (0.047)	0.007 (0.019)	-0.397*** (0.082)	-0.265** (0.117)	
N	4,513	14,063	48,559	95,581	12,031	7,123	39,018
R ²	0.062	0.080	0.075	0.064	0.070	0.078	0.075
Time FE	Y	Y	Y	Y	Y	Y	Y
Country Pair FE	Y	Y	Y	Y	Y	Y	Y
Dominant Country	Australia	China	Euro Area	Other	Russia	United Kingdom	United States

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. The columns restrict the sample to various (destination) country groups. Column (1) includes countries where the dominant country is Australia. Column (2) includes countries where the dominant country is China. Column (3) includes countries where the dominant country is the euro area. Column (4) includes countries where the dominant countries are all others. Column (5) includes countries where the dominant country is Russia. Column (6) includes countries where the dominant country is the United Kingdom. Column (7) includes countries where the dominant country is the United States.

Table 8
Tourist Arrival Regression, by Peg.

	Dependent Variable: Δ Tourist Arrivals		
	(1)	(2)	(3)
Δ USD vs. Origin Country	−0.110*** (0.021)	−0.116*** (0.021)	−0.106*** (0.021)
Δ Bilateral Exchange Rate	−0.073*** (0.014)	−0.074*** (0.014)	−0.075*** (0.015)
Δ Dominant Currency vs. Origin Country	−0.002 (0.015)	−0.008 (0.016)	−0.008 (0.015)
Peg	−0.015* (0.008)	−0.043*** (0.009)	−0.002 (0.009)
Δ USD vs. Origin Country*Peg	0.026 (0.053)	0.071 (0.050)	
Δ Bilateral Exchange Rate*Peg			0.027 (0.048)
Δ Dominant Currency vs. Origin Country*Peg	−0.099 (0.075)	0.018 (0.046)	0.041 (0.049)
N	220,888	220,888	220,888
R ²	0.064	0.064	0.064
Time FE	Y	Y	Y
Country Pair FE	Y	Y	Y
Peg	Bilateral	Dest. USD	Origin USD

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC. Peg is dummy that is one if there is a peg between country i and country j in column (1), there is a peg between the destination country j and the USD in column (2), and whether there is peg between the origin country i and the USD in column (3).

a statistically significant effect for tourist flows in their respective markets. This result again confirms the uniquely important role of the U.S. dollar in international tourism.

Finally, we discuss the role of exchange rate regimes. One may suspect that the results are driven by tourism-dependent countries with pegs, where significant pricing takes place in dollars. In Table 8, we re-estimate our baseline equation, interacting all three exchange rate movements with a dummy that takes the value for various exchange rate regimes: (i) a peg between the destination and origin country, (ii) a peg between the destination country and the U.S. dollar, and (iii) a peg between the origin country and the U.S. dollar. The interaction term illustrates the differential effect of pegged countries relative to non-pegged countries. In all three columns, all interaction effects are statistically insignificant, indicating that pegged countries do not respond differentially to exchange rate movements than non-pegged countries in cases in which we can estimate the differential elasticity.

More specifically, in column (1), we test how tourist flows are differentially affected by exchange rate movements if two countries are pegged to each other—for example, within the euro area. Of course, we cannot test the exchange rate elasticity for bilateral exchange rate movements, as the exchange rate is, by definition, fixed. However, we can test the differential response of an euro-area country relative to a non-euro-area country in response to movements in the U.S. dollar exchange rate or the dominant exchange rate. For instance, in the case of tourist flows to Ireland, we compare the elasticity of German and Canadian tourist flows in response to movements in the U.S. dollar or the British pound (the dominant currency for Ireland) against the euro and the Canadian dollar. We do not find evidence that pegged countries (in this case, Germany) respond differentially to exchange rate movements than non-pegged countries (Canada), as shown by the insignificant interaction terms. In column (2), we define a peg as a dummy that is equal to one if the destination country is pegged against the U.S. dollar, as is—for example—the case for many Caribbean countries. In this case, the bilateral exchange rate movement between any origin country and destination country for which the peg dummy is one is perfectly collinear with the movement in the U.S. dollar. For instance, the movement in the euro against the U.S. dollar is the same as the euro against the Bahamian dollar (which is pegged to the U.S. dollar). This collinearity prevents us from estimating the U.S. dollar exchange rate elasticity separately from the bilateral exchange rate elasticity for this set of countries. However, we can test whether tourism is more or less elastic to U.S. dollar or dominant currency movements toward countries that have a peg with the U.S. dollar. We do not find evidence in favor of a differential elasticity. In column (3), we define a peg as a dummy that is equal to one for an origin country that has a pegged exchange rate to the U.S. dollar (e.g., travel flows from Hong Kong, which has a peg against the U.S. dollar). As for the other peggers, we do not find evidence that these countries exhibit a differential exchange rate elasticity than countries that are not pegged to the U.S. dollar.

7. Domestic tourism

In this section, we analyze how domestic tourism shapes the consequences of exchange rate changes for tourist arrivals and hotel prices. To do so, we interact the share of domestic tourism with the change in exchange rates. Unfortunately, the

data is not available for all countries so our sample shrinks for this exercise. Table A2 shows the results for hotel prices across hotel categories. For all hotels, as shown in column (1), for countries where the share of domestic tourists is low, local currency hotel prices are highly sensitive to changes in the U.S. dollar. The large elasticity suggests that U.S. dollar pricing is more common for hotels in those countries, as their tourism sector depends more on foreign tourists. With a higher share of domestic tourists, the elasticity decreases, indicating that hotels are more likely to price in domestic currency. The results are strongest for medium to high-quality hotels and less strong for low-quality hotels.

Table A3 tests whether tourist arrival elasticities to exchange rate changes are affected by the share of domestic tourists in the destination country. As expected, we find that in countries where domestic tourism is more important, the bilateral exchange rate matters more for tourist arrivals, consistent with the finding for hotel pricing.

8. Conclusion

In this paper, we estimate exchange rates elasticities of international tourism and find that. Contrary to the conventional wisdom that international trade of services is usually invoiced in exporting-country currencies and that a domestic depreciation is beneficial to export volumes, our analysis points out that the benefits of exchange rate flexibility in international tourism are damped by the effect of DCP. The U.S. dollar plays a particularly important role even in countries where the largest share of foreign tourist origins are from non-U.S. countries. To the extent that hotels may choose to invoice their services in the U.S. dollar or other foreign currencies, as indicated by the partial pass-through of dollar exchange rate movements to hotel prices, a general strengthening of the dollar could have a contractionary effect on tourist arrivals for destination countries with strong U.S. dollar pricing. Quantitatively, a 1 percent U.S. dollar appreciation against all other currencies can be associated with a 0.12 percent decline within a year in tourism flows. Our paper thus complemented the DCP literature by providing evidence of DCP in international tourism.

While the extent of DCP varies across countries partly as a result of country-specific characteristics of the tourism industry, in general, there is strong complementarity between DCP and DCF. We show that in countries with higher U.S. dollar borrowing, hotel prices in domestic currency are extremely sensitive to fluctuations in the dollar exchange rate, whereas in countries with low U.S. dollar borrowing, changes in hotel prices are orthogonal to currency movements relative to the U.S. dollar. Moreover, in low U.S. dollar borrowing countries, the bilateral exchange rate dominates the dollar exchange rate in driving tourist flows, which is the opposite for high dollar-borrowing countries. These results are consistent with the literature on DCP and DCF.

Overall, our results indicate that the widespread DCP in the international tourism industry can weaken the response of tourism exports to exchange rate movements. For small tourism-dependent economies, the benefits of exchange rate flexibility may be muted if local hotels choose to invoice services in a foreign currency, either as a hedge against foreign borrowing costs or to match the preferences of foreign tourists. For countries where the U.S. dollar is the common invoicing currency, the tightening of the U.S. monetary policy would imply a strengthening of the dollar over the medium term and an increase in the hotel prices measured in local currencies and currencies of other tourist origin countries. To mitigate such an adverse effect on tourism exports, policymakers may need to consider more supportive macroeconomic policies as local tourism sectors recover from the COVID-19 pandemic. Over the longer run, despite the limited effect of exchange rate flexibility on export volumes due to DCP, countries can still improve competitiveness through structural reforms such as reducing the unit labor costs by enhancing labor market flexibility and improving domestic access to finance, which in turn may reduce the tourism sector's reliance on foreign borrowing and therefore strengthen the benefits of exchange rate flexibility as a shock absorber.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

None.

Appendix

Fig. A5

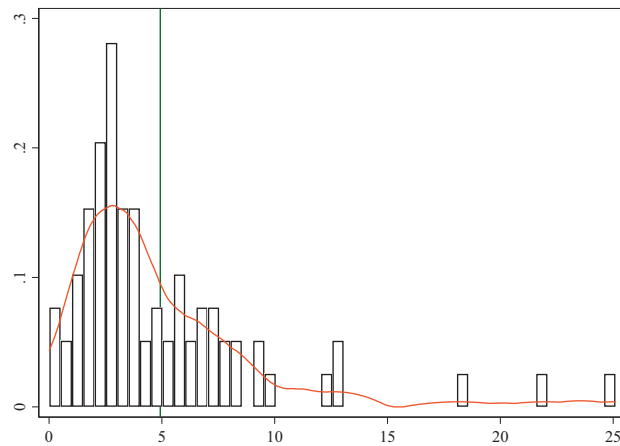


Fig. A1. Tourism Share across Countries. Note: This graph shows the median tourism value added as a share of GDP across countries in 2019. The green line reflects the average.

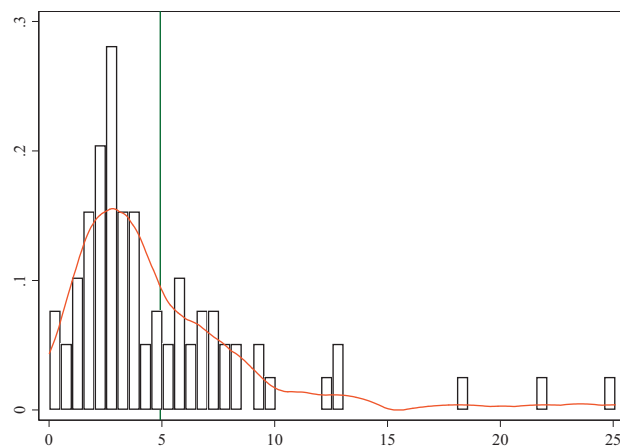


Fig. A2. Tourism Value Added as a share of GDP over Time. Note: This graph shows the median tourism value added as a share of GDP and its interquartile range over time.



Fig. A3. Tourism Population Ratio over Time. Note: This graph shows the median tourist arrivals as a share of the population and its interquartile range over time.

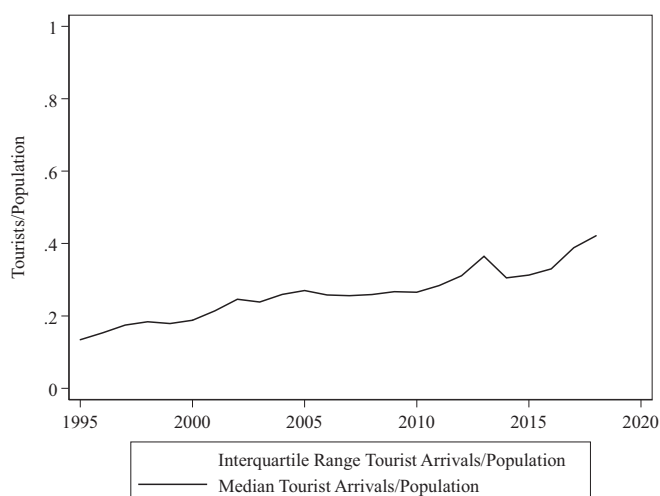


Fig. A4. Hotel Prices over time. Note: This graph shows the median hotel price in U.S. dollars from Tripadvisor and their interquartile range over time.

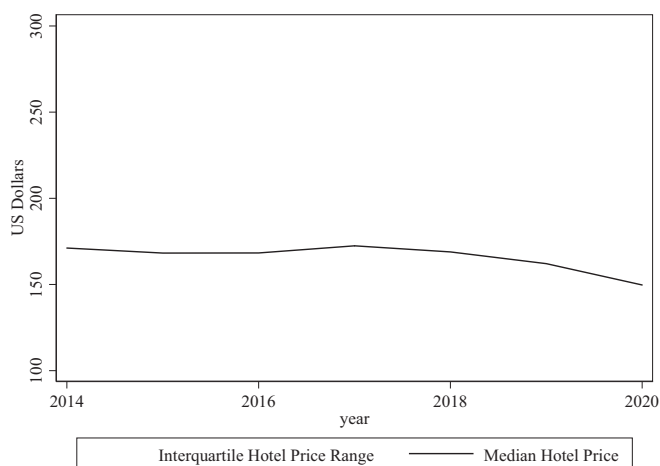


Fig. A5. Hotel Prices for different categories. Note: This graph shows the median hotel price in U.S. dollars from Tripadvisor for different hotel categories averaged across years.

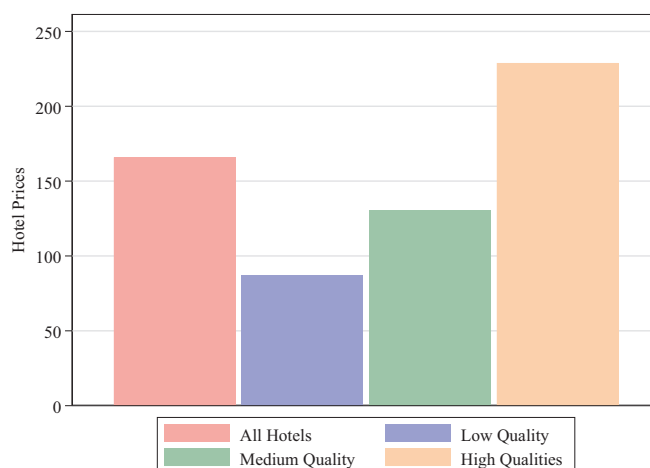


Fig. A6. Dynamic Response of Tourist Arrivals to Exchange Rate Changes. Note: This graph shows the estimated dynamic effects of the percent bilateral depreciation of the origin country relative to the destination country (blue), percent depreciation of the origin country relative to the U.S. dollar (pink) for destination country.

Table A1

Summary Statistics.

	Mean	P25	P75	Std.Dev.	N
Δ Arrivals	0.075	-0.094	0.242	0.456	221,624
Δ Bilateral Exchange Rate	-0.002	-0.049	0.048	0.120	221,624
Δ USD vs. Origin Country	0.027	-0.019	0.054	0.093	221,624
Δ Dominant Currency vs. Origin Country	0.007	-0.029	0.044	0.105	221,624

Note: This table shows the summary statistics for the tourist arrival regressions. Δ Arrivals is change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Δ Dominant Currency vs. Origin Country is the change in the log exchange rate between origin country j relative to (destination) country-specific dominant currency (CSDC), where a positive sign indicates a depreciation of the origin country relative to the (destination) CSDC.

Table A2

Hotel Price Pass-Through Regression and the Share of Domestic Tourism.

	Dependent Variable: Δ Local Hotel Prices			
	(1)	(2)	(3)	(4)
% Depreciation against USD	0.579*** (0.145)	0.461** (0.173)	0.728*** (0.174)	0.436 (0.398)
Share Domestic Tourists	0.003 (0.002)	0.003 (0.004)	0.002 (0.002)	0.007 (0.005)
% Depreciation against USD * Share Domestic Tourists	-0.040* (0.021)	-0.047* (0.024)	-0.047*** (0.015)	-0.031 (0.047)
N	708	708	708	655
R ²	0.183	0.163	0.162	0.128
Hotels	All Hotels	High Quality	Medium Quality	Low Quality
Time FE	Y	Y	Y	Y
Country FE	Y	Y	Y	Y

Note: This table shows the results from a hotel price pass-through regression. The dependent variable is the change in the log average hotel price in local currency of a country i where the hotels are based between year t and $t-1$. *percent Depreciation against USD* is the change between the log exchange rate of country i relative to the U.S. dollar (USD). *Share Domestic Tourists* is the share of domestic tourists in country i . Column (1) includes all hotels. Columns (2), (3), and (4) only includes high, medium, low quality hotels, respectively.

Table A3

Tourist Arrivals and Share of Domestic Tourism.

	Dependent Variable: Δ Arrivals			
	(1)	(2)	(3)	(4)
Δ Bilateral Exchange Rate	-0.102*** (0.025)		-0.018 (0.035)	0.069 (0.053)
Share Domestic Tourists	-0.026*** (0.004)	-0.030*** (0.004)	-0.026*** (0.004)	-0.038*** (0.008)
Δ Bilateral Exchange Rate*Share Domestic Tourists	-0.022** (0.011)		-0.039** (0.017)	-0.049** (0.021)
Δ USD vs. Origin Country		-0.205*** (0.032)	-0.176*** (0.044)	-0.239*** (0.061)
Δ USD vs. Origin Country*Share Domestic Tourists		0.010 (0.012)	0.037** (0.017)	0.027 (0.020)
N	76,972	77,694	76,589	52,260
R ²	0.089	0.088	0.090	0.111
Time FE	Y	Y	Y	Y
Country Pair FE	Y	Y	Y	Y

Note: This table shows the results from a tourist arrival regression. The dependent variable is the change in the log number of tourists arriving from country j to country i between year t and $t-1$. Δ Bilateral Exchange Rate is the change in the log exchange rate between origin country j relative to destination country i , where a positive sign indicates a depreciation (appreciation) of the origin (destination) country. Δ USD vs. Origin Country is the change in the log exchange rate between origin country j relative to the U.S. dollar (USD), where a positive sign indicates a depreciation of the origin country relative to the USD. Share Domestic Tourists is the share of domestic tourists in the destination country. Time (year) and country-pair fixed effects are included in all columns.

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